

Evidence Against Equilibrium: Optimal Rates and Anytime-Valid Tests of Detailed Balance

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Abstract

Whether a stationary system is at equilibrium or driven out of it can be read from a single trajectory: driven Markov dynamics break detailed balance and look different when played backward. In practice such data are examined as they arrive, and a classical likelihood-ratio test that is correct at a planned sample size stops being correct when checked repeatedly. In our simulations its false-alarm rate rose from 5% to 24–60%. We develop tests of detailed balance that remain valid when checked after every observation, and we characterize how fast any such test can succeed. Against the composite null hypothesis of reversibility, no test can accumulate evidence faster than $\rho = \sum_{ij} f_{ij} \log_2 \frac{2f_{ij}}{f_{ij}+f_{ji}}$ bits per step, where f_{ij} is the stationary edge flow. This is the divergence to the nearest reversible chain, not the entropy production σ : $\rho \leq \sigma/2$, $\rho \approx \sigma/4$ near equilibrium, and ρ is finite when σ is infinite. The rate yields a lower bound of about $\log_2(1/\alpha)/\rho$ on expected detection time at level α . A likelihood e-process based on universal inference attains ρ asymptotically but is slow at practical α . A second test, which bets on the orientation of loops through a reference state, is exactly valid under continuous monitoring. Near equilibrium it captures 92–100% of the optimal rate and needs 1.7–1.9 times the sample size of an idealized fixed-sample test. Results were checked by an independent adversarial review and reproduced independently.

1 Introduction

A stationary Markov chain with transition matrix T and stationary law π is *reversible*, or satisfies detailed balance, if $\pi_i T_{ij} = \pi_j T_{ji}$ for all states i, j . Reversibility is the signature of thermodynamic equilibrium. Its violation, broken detailed balance, signals that the system is driven, for example by ATP hydrolysis in molecular motors or by external forcing in active matter. The entropy production rate σ quantifies the violation, and a substantial literature estimates it from observed trajectories (Section 2).

This paper concerns a prior question: *deciding* whether detailed balance is broken, as data arrive. Experimenters rarely fix a recording length in advance and look once. They watch, and stop when the evidence looks convincing. That practice invalidates classical tests. In our simulations of reversible chains (Section 7), a likelihood-ratio test calibrated to 5% at a fixed sample size raised false alarms 24–36% of the time when checked 10 times and 44–60% of the time when checked continuously.

We make four contributions.

1. **The optimal detection rate** (Section 4). Against the composite null “some reversible chain,” the expected log-evidence of any valid test grows at most at rate ρ , the Kullback–Leibler rate from T to its nearest reversible chain $(T + T^*)/2$. We show $\rho \leq \sigma/2$, $\rho \approx \sigma/4$ near equilibrium, and $\rho < \infty$ even when $\sigma = \infty$. A Wald-type argument gives $\mathbb{E}[\tau] \gtrsim \log_2(1/\alpha)/\rho$ for every level- α sequential test.
2. **A rate-optimal e-process** (Section 5). A running maximum-likelihood construction [24] gives an e-process whose evidence grows at exactly rate ρ . It pays a model-selection overhead of order $(k(k-1)/2) \log t$ bits, which dominates at practical significance levels.
3. **The loop test** (Section 6). Under detailed balance, each loop through a reference state is as likely as its reversal, so its orientation is a fair coin. Betting on orientations gives a test that is exactly valid under continuous monitoring, needs no constrained optimization, and pays little overhead. Near equilibrium, where detection is hard, it captures 92–100% of the optimal rate.
4. **Experiments** (Section 7) confirm validity under monitoring after every step, the predicted growth rates, and detection times within 1.7–1.9 times an idealized fixed-sample test near equilibrium.

The paper supersedes an earlier version organized around “evidence holonomy,” a framing of irreversibility as the change in description length around a loop of representation transforms. That framing and the entropy-production estimator it motivated are retained in the appendices. The present results can be read as its sharpest form: evidence accumulated around the loops of the chain itself.

2 Related work

Entropy production and time reversal. For stationary Markov processes, the relative-entropy rate between forward and time-reversed path measures equals the entropy production rate [20, 6, 21]. Roldán and Parrondo estimated this divergence from single trajectories with plug-in and compression-based estimators, the latter via Ziv–Merhav cross-parsing [18, 19, 26]. Chazottes and Redig [2] and Cristadoro et al. [3] built universal estimators from hitting and waiting times. Coarse-grained or partial observation yields lower bounds on dissipation [7], and waiting-time statistics can reveal irreversibility that transition counts miss [12]. Compression [13, 16] and neural networks [10] have also been used to estimate entropy production. Cycle counts and affinities in Markov models are analyzed in [14].

Deciding the arrow of time. Roldán et al. [17] apply Wald’s sequential probability ratio test to decide the direction of time’s arrow. Their test compares a process with its own time reversal, a *simple* null, and evidence accrues at rate σ . We test the *composite* null “some reversible chain,” which asks whether the system is at equilibrium at all, without specifying which equilibrium. Section 4 shows the relevant rate is then $\rho \leq \sigma/2$.

Testing reversibility. Fixed-sample tests of reversibility for Markov chain data include chi-squared tests on transition counts [22] and algebraic checks [9]. Property testing studies identity testing of reversible chains [5]. The reversible chains form both an exponential and a mixture family; the m-projection of a kernel onto them is the additive reversibilization $(T + T^*)/2$ [25, 4], identified as the closest equilibrium dynamics in [1]. Our Lemma 4.1 restates this result.

Anytime-valid inference. E-values and e-processes give tests that stay valid under optional stopping and continuation [8, 15]. Universal inference constructs them for composite nulls from a predictable likelihood divided by the null maximum likelihood [24]. To our knowledge, neither the rate ρ as the optimal evidence rate against reversibility nor anytime-valid tests of detailed balance have appeared before. Our literature search was, however, limited.

3 Setup

Let $x_0, x_1, \dots$ be a Markov chain on $\mathcal{X} = \{1, \dots, k\}$ with transition matrix T . When T is irreducible, π denotes its stationary law, $f_{ij} = \pi_i T_{ij}$ the stationary edge flow, and $T_{ij}^* = \pi_j T_{ji}/\pi_i$ the time reversal. The entropy production rate is

$$\sigma = \sum_{i,j} f_{ij} \log_2 \frac{f_{ij}}{f_{ji}} \quad (\text{bits/step}),$$

equal to $+\infty$ if some transition is possible in only one direction [6]. The null hypothesis is

$$\mathcal{H}_0 : T \text{ is reversible (detailed balance holds), with any initial state.}$$

A *level- α anytime-valid test* is a process (E_t) , adapted to the data, such that $P(\exists t : E_t \geq 1/\alpha) \leq \alpha$ for every law in $\mathcal{H}_0$. It may be checked after every observation. An *e-process* satisfies the stronger condition $\mathbb{E}[E_\tau] \leq 1$ for every stopping time τ and every null law. Every nonnegative martingale with initial value 1 is an e-process, and Ville's inequality makes it a level- α test. Logarithms are base 2 unless stated.

4 The optimal rate

Lemma 4.1 (Nearest reversible chain [25]). *Let T be irreducible. Among reversible transition matrices Q , the rate $D_\pi(T\|Q) = \sum_{i,j} f_{ij} \log_2(T_{ij}/Q_{ij})$ is minimized by the additive reversibilization $M = (T + T^*)/2$, and*

$$\min_{Q \text{ reversible}} D_\pi(T\|Q) = \rho := \sum_{i,j} f_{ij} \log_2 \frac{2f_{ij}}{f_{ij} + f_{ji}}.$$

Proof. Let g be the symmetric edge measure of a reversible Q , with marginal μ . Then $D_\pi(T\|Q) = D(f\|g) - D(\pi\|\mu)$. Let $s = (f + f^\top)/2$, which is symmetric with marginal π by stationarity. Since $\log(s/g)$ is symmetric, $\sum f \log(s/g) = \sum s \log(s/g)$, hence $D(f\|g) = D(f\|s) + D(s\|g)$. By data processing, $D(s\|g) \geq D(\pi\|\mu)$. So $D_\pi(T\|Q) \geq D(f\|s) = \rho$, with equality when $g = s$, that is, $Q = M$. $\square$

Proposition 4.2 (Properties of ρ). (a) $0 \leq \rho \leq \sigma/2$, and $\rho = 0$ if and only if T is reversible.

(b) $\rho \leq \sum_{i \neq j} f_{ij}$ bits. In particular, $\rho < \infty$ even when $\sigma = \infty$.

(c) If $f_{ij} = s_{ij} + a_{ij}$ and $f_{ji} = s_{ij} - a_{ij}$ with $|a_{ij}| \ll s_{ij}$, then $\rho = \sigma/4 + O(\max |a|^3/s^2)$.

Proof. (a) By convexity of D in its second argument, $D(f\|\frac{1}{2}f + \frac{1}{2}f^\top) \leq \frac{1}{2}D(f\|f) + \frac{1}{2}D(f\|f^\top) = \sigma/2$. (b) Each term satisfies $\log_2(2f_{ij}/(f_{ij} + f_{ji})) \leq 1$, and diagonal terms vanish. (c) For each unordered pair, ρ contributes $(s + a) \log \frac{s+a}{s} + (s - a) \log \frac{s-a}{s} = a^2/s + O(a^4/s^3)$ nats, while σ contributes $2a \log \frac{s+a}{s-a} = 4a^2/s + O(a^4/s^3)$ nats. $\square$

Theorem 4.3 (No test can do better). *Let T be irreducible and irreversible, and let the chain start from π . Let P denote its law.*

- (a) *For every e-process (E_t) valid under $\mathcal{H}_0$, $\mathbb{E}_P[\log_2 E_t] \leq t\rho$ for all t .*
- (b) *Every sequential test with type-I error at most α under $\mathcal{H}_0$, power at least $1 - \beta$ against P , and $\mathbb{E}_P[\tau] < \infty$ satisfies*

$$\mathbb{E}_P[\tau] \geq \frac{d(1 - \beta \| \alpha) - c_T}{\rho},$$

where $d(p \| q)$ is the binary KL divergence in bits and c_T depends only on T . As $\alpha, \beta \rightarrow 0$, the bound is $(1 + o(1)) \log_2(1/\alpha)/\rho$.

Proof. The chain with transition matrix $M = (T + T^*)/2$ started from π is reversible and stationary, so its law P^M belongs to $\mathcal{H}_0$. Moreover, M has stationary law π and the support of T , so $D(P_t \| P_t^M) = t\rho$ exactly.

- (a) By Gibbs' inequality, $\mathbb{E}_P[\log E_t] \leq D(P_t \| P_t^M) + \log \mathbb{E}_{P^M}[E_t] \leq t\rho$.
- (b) By data processing on the stopped path, $D(P_\tau \| P_\tau^M) \geq d(1 - \beta \| \alpha)$. The log-likelihood ratio $\sum_{s \leq \tau} \log_2 \frac{T(x_{s-1}, x_s)}{M(x_{s-1}, x_s)}$ is a Markov additive functional with stationary mean ρ per step. By Wald's identity with the Poisson-equation correction, whose bounded remainder defines c_T , $D(P_\tau \| P_\tau^M) \leq \rho \mathbb{E}_P[\tau] + c_T$. $\square$

Remark 4.4 (Why ρ and not σ). The simple-null test of [17] compares P with its time reversal P^* and accrues evidence at rate $D(P \| P^*) = \sigma$. Here the least favorable member of the null is M , not P^* , and $D(P \| M) = \rho \leq \sigma/2$. Deciding whether a system is at equilibrium, without knowing which equilibrium, is intrinsically slower than deciding which way time runs.

5 A rate-optimal likelihood e-process

Let $C_{ij}(t)$ count the transitions $i \rightarrow j$ among the first t , and let $q_{s-1}(j | i) = (C_{ij}(s-1) + \frac{1}{2}) / (\sum_j C_{ij}(s-1) + \frac{k}{2})$ be the Krichevsky–Trofimov (KT) predictor [11]. Define

$$E_t = \frac{\prod_{s=1}^t q_{s-1}(x_s | x_{s-1})}{\sup_{Q \text{ reversible}} \prod_{s=1}^t Q(x_{s-1}, x_s)}.$$

The denominator is the reversible maximum likelihood. We compute it with the standard self-consistent iteration $X_{ij} \leftarrow (C_{ij} + C_{ji}) / (N_i/x_i + N_j/x_j)$, with $Q_{ij} = X_{ij}/x_i$ [23].

Proposition 5.1 (Validity). *Under $\mathcal{H}_0$, (E_t) is an e-process. In particular, $P(\exists t : E_t \geq 1/\alpha) \leq \alpha$ for any initial state.*

Proof. If T_0 is the true reversible matrix, the denominator is at least $\prod_s T_0(x_{s-1}, x_s)$. So $E_t \leq \prod_s q_{s-1}(x_s | x_{s-1}) / T_0(x_{s-1}, x_s)$, which is a nonnegative martingale with initial value 1 [24]. $\square$

Proposition 5.2 (Growth). *If T is irreducible, $\frac{1}{t} \log_2 E_t \rightarrow \rho$ almost surely.*

Proof sketch. The KT numerator is within $\frac{k(k-1)}{2} \log_2 t + O(k)$ bits of the unconstrained maximum likelihood. The empirical edge frequencies converge to f . The unconstrained and reversible maximized log-likelihoods, divided by t , converge to $\sum f \log T$ and $\max_{Q \text{ rev}} \sum f \log Q$; the second step uses continuity of the reversible maximum in the frequencies. Their difference is ρ by Lemma 4.1. $\square$

By Theorem 4.3(a), (E_t) is growth-rate optimal. At finite t , however, its log-evidence is roughly $\rho t - \frac{k(k-1)}{2} \log_2 t$. The numerator pays for predicting every transition probability, while the denominator fits the symmetric part for free. At $\alpha = 0.05$ this overhead dominates. The heuristic detection time, the solution of $\rho\tau = \log_2(1/\alpha) + \frac{k(k-1)}{2} \log_2 \tau$, overpredicts the observed medians by a factor of 1.2–1.9, but it identifies the source of the cost.

Numerical note. The iteration is stopped by a relative-change rule, not a certified optimality gap. In an independent check against multi-start quasi-Newton optimization on 150 random count matrices, it fell short of the maximum by at most 1.5×10^{-4} nats, which inflates E_t by a factor of at most about 1.00015.

6 The loop test

Construction. Count time in transitions. Let $B = 10k$ and let r be the most frequent state among $x_0, \dots, x_B$, with ties going to the lowest index. From the first time $t \geq B$ with $x_t = r$, cut the path into *loops* $e = (r, y_1, \dots, y_{m-1}, r)$ that leave r and return to it. Let $\bar{e}$ denote the reversed loop. When a loop that started at time t_0 closes, multiply E (initially 1) by

$$\frac{2q(e)}{q(e) + q(\bar{e})}, \quad q(e) = \prod_s q(y_{s+1} \mid y_s), \quad q(\bar{e}) = \prod_s q(y_s \mid y_{s+1}),$$

where q is the KT estimate from all transitions $1, \dots, t_0$, frozen at the loop's start. Palindromic loops give a factor of exactly 1. Reject reversibility the first time $E \geq 1/\alpha$.

Theorem 6.1 (Validity of the loop test). *Under $\mathcal{H}_0$, $P(\exists t : E_t \geq 1/\alpha) \leq \alpha$.*

Proof. By the strong Markov property, each loop is independent of the past at its start, including the choice of r (which uses only $x_0, \dots, x_B$) and the frozen estimate. Under detailed balance, $P(e) = \prod T = \prod T^\top$ along any closed walk (Kolmogorov's criterion), so $P(e) = P(\bar{e})$. Given the unordered pair $\{e, \bar{e}\}$ and the past, the observed orientation is therefore a fair coin, and each factor has conditional mean $\frac{q(e)}{q(e) + q(\bar{e})} + \frac{q(\bar{e})}{q(e) + q(\bar{e})} = 1$. Indexed by completed loops, E is a nonnegative martingale with initial value 1. It is constant between completions, so its supremum over all steps equals its supremum over completions. Ville's inequality gives the claim. $\square$

Remark 6.2 (What is not claimed). (E_t) is not an e-process with respect to step-level stopping times. A stopping rule that inspects a partially completed loop can make $\mathbb{E}[E_\tau]$ exceed 1; the review exhibited $\mathbb{E}[E_\tau] \approx 1.18$. The guarantee is Theorem 6.1: a level- α test that may be checked after every step. For the same reason, averaging the processes for different reference states is not covered, because they evolve on different clocks.

Efficiency. With oracle knowledge of T , the evidence per loop is $\mathbb{E}[\log_2(2P(e)/(P(e) + P(\bar{e})))]$. Divided by the mean loop length, this is at most ρ by Theorem 4.3(a). Near equilibrium almost all the evidence is carried by loop orientations, and the test is nearly optimal. Far from equilibrium it discards within-loop information, but such chains are detected within about a hundred steps (Table 1).

Why loops. Kolmogorov's criterion states that a chain is reversible exactly when every cycle has equal forward and backward probability, that is, zero cycle affinity [14]. The loop test turns this criterion into a betting game on the cycles the trajectory actually traverses. Evidence accumulates only where time-reversal symmetry could be broken, which is the property Section 5's likelihood

Table 1: Oracle evidence rate of the loop test with the reference state it selects, relative to the optimal rate ρ (bits/step). Simulation of up to 6×10^5 loops per chain; $\pm$ one standard error.

Chain	ρ (bits/step)	Loop-test rate	Efficiency
ring p=0.3 q=0.25	0.0033	0.0033 ± 0.0001	1.00 ± 0.02
ring p=0.3 q=0.2	0.0145	0.0136 ± 0.0003	0.94 ± 0.02
random k=4 #0	0.0189	0.0174 ± 0.0004	0.92 ± 0.02
ring p=0.3 q=0.12	0.0575	0.0458 ± 0.0009	0.80 ± 0.02
random k=4 #1	0.0599	0.0480 ± 0.0010	0.80 ± 0.02
ring p=0.3 q=0.05	0.1429	0.0809 ± 0.0009	0.57 ± 0.01
random k=4 #2	0.1937	0.1116 ± 0.0015	0.58 ± 0.01
one-way ring p=0.3	0.3000	0.1014 ± 0.0006	0.34 ± 0.00

e-process lacks. Chains whose transition graph is a tree are always reversible, and on such graphs every factor is exactly 1.

7 Experiments

All experiments are seeded; `scripts/sequential_experiments.py` regenerates the ancillary files in `anc/`, and `scripts/sequential_tables.py` and `scripts/plot_sequential.py` regenerate the tables and figures. Throughout, $\alpha = 0.05$ and chains start from their stationary law.

Validity. Table 2 and Figure 1 report false-alarm rates over 1,000 trajectories of 20,000 steps for each reversible setting. The loop test is checked after every step. The likelihood e-process and the likelihood-ratio test (LRT) are checked at about 300 checkpoints: every step up to 64, then a geometric grid. The LRT uses the χ^2 reference distribution with $(k-1)(k-2)/2$ degrees of freedom, the number of independent cycle constraints on a complete graph.

Both anytime tests stay within level. The likelihood e-process raised no false alarms in 8,000 runs, which is consistent with its conservative overhead. The LRT is calibrated at its planned sample size (4.2–5.7%) but rejects 24–36% of reversible trajectories when checked 10 times and 44–60% when checked at every checkpoint. The birth–death chain lives on a tree and is always reversible; every loop factor there is exactly 1.

Three settings observe a reversible hidden chain through a noisy or lumping channel. The observed processes are reversible but not Markov, so no theorem above covers them. The loop test’s false-alarm rate there is 2.6–4.5%, which is within level. This is evidence, not a guarantee.

Table 2: False-alarm rates on reversible processes at $\alpha = 0.05$ (1,000 trajectories of 20,000 steps; 95% Clopper–Pearson intervals for the loop test). “Hidden” rows are reversible but not Markov.

Reversible process	Type	Loop test	Likelihood	LRT, fixed n	LRT, 10 looks	LRT, all looks
random k=3	Markov	3.7% [2.6, 5.1]	0.0%	5.7%	30.1%	53.2%
random k=4	Markov	3.5% [2.4, 4.8]	0.0%	4.4%	30.4%	51.1%
random k=5	Markov	3.1% [2.1, 4.4]	0.0%	5.5%	35.5%	59.8%
birth-death k=5	Markov	0.0% [0.0, 0.4]	0.0%	n/a	n/a	n/a
unbiased ring k=3	Markov	2.6% [1.7, 3.8]	0.0%	5.5%	29.1%	49.4%
reversible k=4, noisy view	hidden	4.5% [3.3, 6.0]	0.0%	4.7%	28.7%	47.9%
reversible k=4, states 2,3 merged	hidden	2.6% [1.7, 3.8]	0.0%	4.7%	28.6%	48.3%
unbiased ring, noisy view	hidden	3.1% [2.1, 4.4]	0.0%	4.2%	23.9%	44.5%

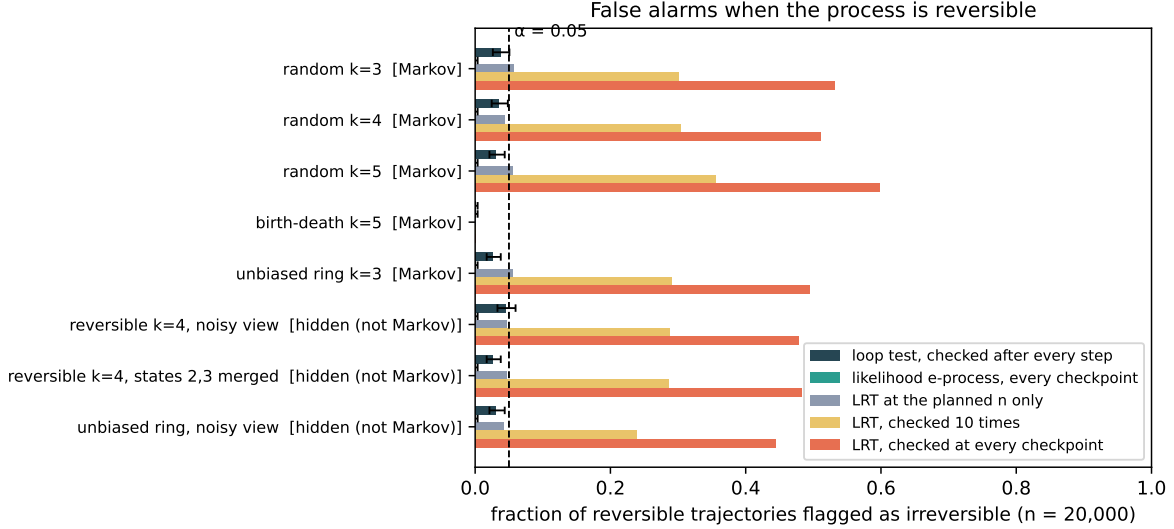


Figure 1: False alarms on reversible processes. The loop test is checked after every step; the likelihood-based tests at about 300 checkpoints.

Detection. Table 3 and Figure 2 report detection times over 400 trajectories per irreversible chain. The comparison is a fixed- n LRT with critical values calibrated by simulating 1,000 trajectories of the least favorable reversible chain M . Its sample size is the smallest n with 80% power against the true chain. This baseline is idealized: it is told the right sample size in advance, which no experimenter knows. With asymptotic χ^2 cutoffs it would appear up to 36% cheaper, because those cutoffs are anti-conservative at small n .

Near equilibrium ($\rho < 0.07$ bits/step), the loop test’s 80th-percentile detection time is 1.7–1.9 times the fixed-sample size, and its median roughly equals that size. Far from equilibrium the ratio rises to 2.5–4.2, because of the burn-in and the discarded within-loop information. These chains are detected within about a hundred steps. The likelihood e-process needs 3.5–8.7 times the fixed-sample size. No median falls below the floor $\log_2(20)/\rho$ of Theorem 4.3. Medians and percentiles carry Monte Carlo error of about $\pm 8\%$.

Table 3: Detection times (steps) on irreversible chains at $\alpha = 0.05$: the fixed- n LRT sample size for 80% power (calibrated by simulation), the loop test’s median and 80th percentile, and the likelihood e-process’s 80th percentile. Ratios are relative to the fixed- n LRT. The floor $\log_2(20)/\rho$ is the asymptotic bound of Theorem 4.3.

Chain	ρ	$\log_2(20)/\rho$	Fixed- n LRT	Loop test (median / 80%)	Likelihood (80%)
ring p=0.3 q=0.25	0.0033	1316	1818	1907 / 3308 (1.8 $\times$)	15686 (8.6 $\times$)
ring p=0.3 q=0.2	0.0145	298	417	407 / 701 (1.7 $\times$)	2834 (6.8 $\times$)
random k=4 #0	0.0189	229	405	436 / 789 (1.9 $\times$)	3539 (8.7 $\times$)
ring p=0.3 q=0.12	0.0575	75	101	123 / 190 (1.9 $\times$)	557 (5.5 $\times$)
random k=4 #1	0.0599	72	119	146 / 226 (1.9 $\times$)	913 (7.7 $\times$)
ring p=0.3 q=0.05	0.1429	30	37	91 / 110 (3.0 $\times$)	184 (5.0 $\times$)
random k=4 #2	0.1937	22	39	83 / 99 (2.5 $\times$)	198 (5.1 $\times$)
one-way ring p=0.3	0.3000	14	22	82 / 93 (4.2 $\times$)	77 (3.5 $\times$)

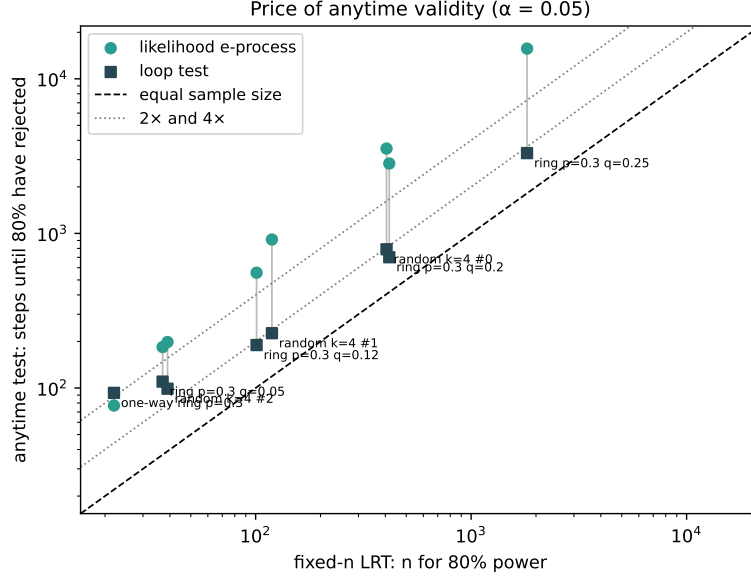


Figure 2: Price of anytime validity: steps until 80% of trajectories have rejected, against the idealized fixed-sample LRT sample size.

Growth rates. Over 20 trajectories of 2×10^5 steps, the likelihood e-process's $\frac{1}{t} \log_2 E_t$ converges to ρ (Figure 3). The final rates and ρ are 0.1430 vs 0.1429, 0.0143 vs 0.0145, 0.2997 vs 0.3000 for a one-way ring with $\sigma = \infty$, and 0.1605 vs 0.1614. The loop test settles at its lower oracle rate.

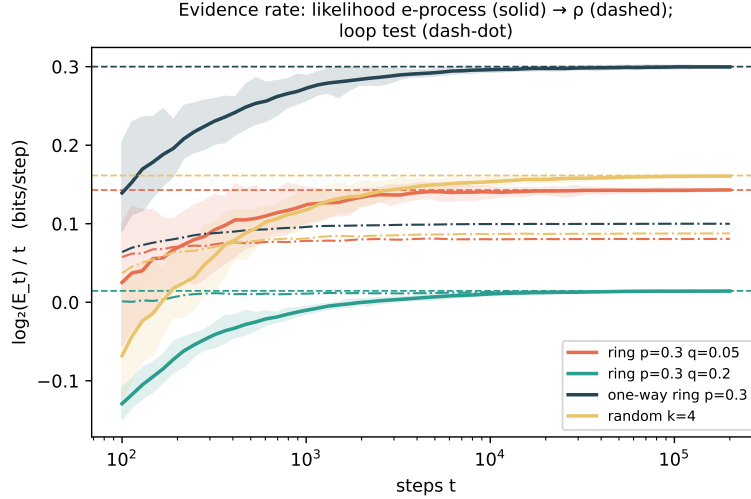


Figure 3: Evidence growth. Solid: likelihood e-process (10–90% band); dashed: ρ ; dash-dot: loop test.

ρ versus σ . Over 300 random chains, ρ lies below $\sigma/2$ and tracks $\sigma/4$ near equilibrium, bending further below it as σ grows (Figure 4).

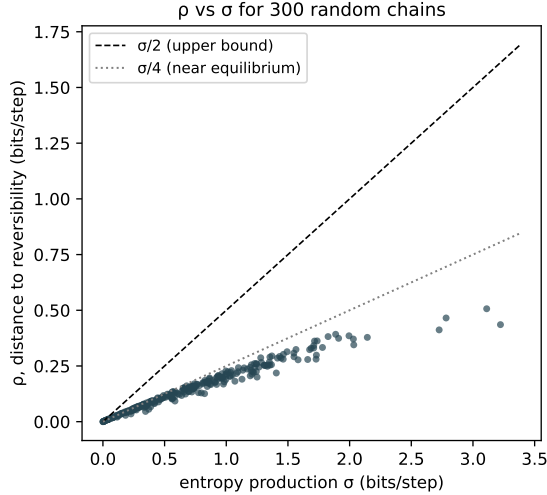


Figure 4: The optimal detection rate ρ against entropy production σ for 300 random chains.

Review and reproduction. Before the final runs, an independent adversarial review checked the proofs and code. It found a gap in an earlier version, which averaged loop tests over reference states; that version was replaced. It also found that the baseline LRT was anti-conservative, and it is now calibrated by simulation. An independent reproduction regenerated the growth, efficiency, and rate experiments with byte-identical output. It also reimplemented the loop test from the written specification alone and obtained 3.8% [2.5, 5.7] false alarms on a reversible four-state chain and a median detection time of 394 steps against our 407 for the $q = 0.2$ ring.

8 Discussion

Summary. Deciding whether a system is at equilibrium, from data examined as they arrive, is governed by ρ , the divergence to the nearest reversible chain, rather than by the entropy production σ . The loop test makes this decision with an exact false-alarm guarantee under continuous monitoring, at a modest cost over an idealized fixed-sample design, and with a one-paragraph proof.

Limitations.

- The theory covers fully observed, finite-state, first-order Markov chains. For hidden-state records the tests stayed within level in our simulations, but the guarantees do not apply. Excursions of an observed state need not regenerate the hidden state.
- The tests certify that detailed balance is broken; they do not give a confidence bound on σ . Because $\sigma \geq 2\rho$, a time-uniform lower confidence bound on ρ would bound σ from below. Inverting the loop test over tilted nulls is one route to it.
- Far from equilibrium the loop test discards within-loop information. Combining it with the likelihood e-process would require a common filtration; Theorem 6.1’s loop-indexed martingale does not provide one.

Extensions. Natural next steps are loop tests for hidden Markov models through regeneration states, higher-order and continuous-time versions, and time-uniform confidence sequences for ρ .

Code and data

Code, ancillary data, and scripts are at <https://github.com/jbwinters/Evidence-Holonomy>. `scripts/reproduce_all.sh` regenerates every number, table, and figure, and this document.

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A Estimating entropy production

The earlier version of this work studied an estimator of σ itself. We summarize it here as a companion to the tests.

The plug-in estimator. Given $x_{0:n-1}$, let $y = x_{1:n-1}$ and $\tilde{y}$ its reversal. Fit KT mixtures over Markov orders $0, \dots, R$ to y and to $\tilde{y}$, freeze them as predictors $\hat{p}$ and $\hat{q}$, and set

$$\hat{\sigma}_{n,R} = \frac{1}{n-1} \sum_t \log_2 \frac{\hat{p}(y_t \mid y_{t-R:t-1})}{\hat{q}(y_t \mid y_{t-R:t-1})}.$$

This is an in-sample plug-in estimate of the divergence of [18, 19]. For an irreducible chain with $T_{ij} > 0 \Leftrightarrow T_{ji} > 0$ and $R \geq 1$, $\hat{\sigma}_{n,R} \rightarrow \sigma$ almost surely. Block frequencies of y and $\tilde{y}$ converge to those of P and P^* , the mixture posterior abandons the inadequate order 0, and the ergodic theorem applies. Estimates for different $R \geq 1$ nearly coincide. That reflects a shared, correctly specified model class, not independence from the choice of coder.

Why universality alone does not justify KL estimation. Pointwise universality of a code controls code lengths only on paths typical for the law it is universal for. Distinct stationary ergodic laws are mutually singular, so a universal code for P^* says nothing about the code length of a path drawn from P . Differences of separate compressions estimate the entropy-rate difference, which vanishes for time reversal. Estimating $D(P\|P^*)$ requires a model of P^* that is accurate on contexts typical for P , and a finite-order plug-in supplies one for finite-state Markov chains.

Hidden-state records. If Y observes a Markov chain Z through a memoryless channel, then reversing the observation equals observing the reversal. By data processing, $D(P_m^Y \| P_m^{Y*}) \leq (m-1)\sigma_Z$ for every m [7]. The order- R estimator targets the conditional increment $d_R = D_{R+1} - D_R$. Averages of these increments are bounded by σ_Z , but a single d_R need not be.

Numerical validation. Table 4 and Figure 5 follow 15 random chains (σ from 0.0067 to 1.92 bits/step). The median relative error falls from 10.3% at $n = 2^9$ to 1.3% at $n = 2^{15}$. On 60 further chains at $n = 2^{15}$, the correlation with σ is 0.9994 (Figure 6). On reversible chains the positive in-sample bias shrinks roughly as $1/n$: 0.0015 bits/step at $n = 2^{11}$ and 0.00013 at 2^{15} for three states. A hidden three-state ring with $\sigma_Z = 0.646$ seen through a noisy channel yields 0.085–0.127 bits/step. Merging two of its states makes the observed process exactly reversible, and the estimate falls to at most 7×10^{-5} (Figure 7).

Table 4: Absolute relative error $|\hat{\sigma} - \sigma|/\sigma$ of the time-reversal estimate ($R = 3$) on 15 random biased chains (five each with 3, 4, and 5 states), by sample length n . Every row uses the same 15 chains.

n	median	interquartile range	max
2^9	10.3%	[2.2%, 14.1%]	123%
2^{11}	8.8%	[5.3%, 11.8%]	86%
2^{13}	4.4%	[3.7%, 7.7%]	31%
2^{15}	1.3%	[0.3%, 3.9%]	19%

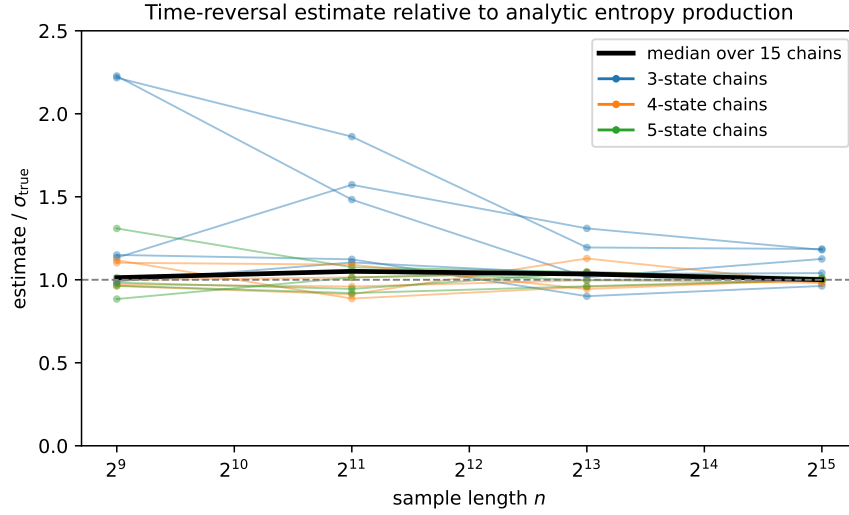


Figure 5: Estimate divided by σ for 15 random chains and their median, as the sample length grows.

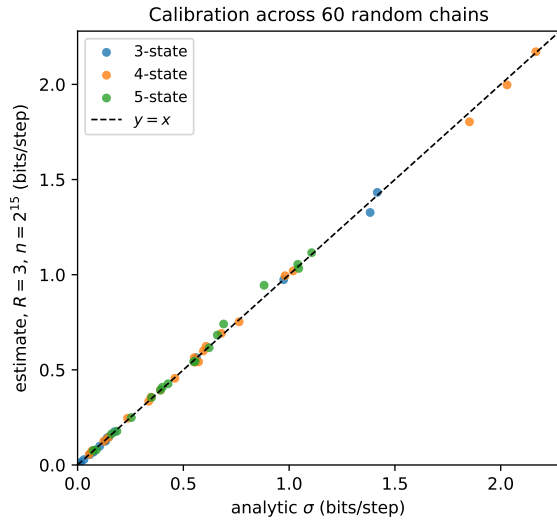


Figure 6: Estimate ($R = 3, n = 2^{15}$) against σ for 60 random chains.

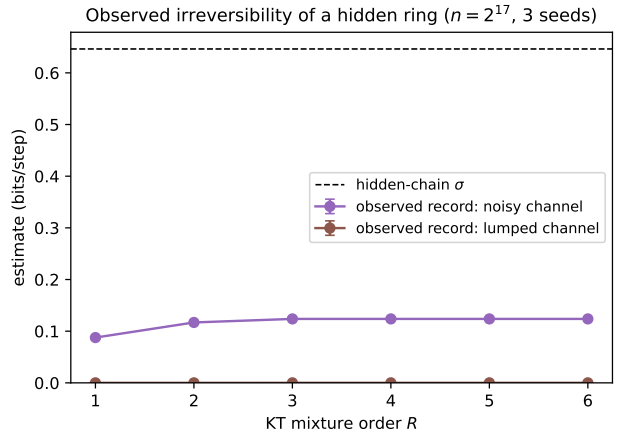


Figure 7: Estimates on observed records of a hidden ring, by mixture order.

B Evidence holonomy

The earlier version framed irreversibility as a holonomy of evidence. A loop of representation transforms L_n maps a path back to the original alphabet. Two functionals compare evidence before and after the loop.

- *Representation-space holonomy* $\mathcal{E}(L_n(x)) - \mathcal{E}(x)$ scores each representation with its own universal code. Its rate is the entropy-rate difference $h(Q) - h(P)$, where $Q = (L_n)\#P$.
- *KL holonomy* $\log_2(P_n(x)/Q_n(x))$ scores the original path under the loop’s pushforward law. Its rate is D-rate $d(P\|Q)$.

For the time-reversal loop, the first vanishes, since reversal preserves entropy rates. The second equals σ for Markov chains [6].

In practice every loop we examined reduced to one of three cases:

- literal reversal, which gives the estimator of Appendix A;
- a bijection, which gives zero;
- a deterministic coarse-grain-and-lift, which gives an infinite KL rate.

So the general formalism organizes known results rather than producing new ones. The loop test of Section 6 is the version of this idea that does work: evidence accumulated around the chain’s own cycles, weighed against their reversals.

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